TEST PLAN

Table of Contents

INTRODUCTION **3**

1.1 OBJECTIVES 3
1.2 TEAM MEMBERS 3

2 SCOPE **3**

3 ASSUMPTIONS / RISKS **7**

3.1 ASSUMPTIONS 7
3.2 RISKS 7

4 TEST APPROACH **7**

4.1 TEST AUTOMATION 8

5 TEST ENVIRONMENT **8**

6 MILESTONES / DELIVERABLES **8**

6.1 TEST SCHEDULE 8
6.2 DELIVERABLES 9

Introduction

The Test Plan has been created to communicate the test approach to team members. It includes the objectives, scope, schedule, risks and approach. This document will clearly identify what the test deliverables will be and what is deemed in and out of scope.

1.1 Objectives

CyberQuest is a web-based cybersecurity awareness training application. It was designed to help employees and students learn how to identify common social engineering and cybersecurity threats.

The purpose of testing is to ensure that CyberQuest functions correctly, provides accurate user feedback, records user progress, and provides a functional experience for users completing the scenarios.

The testing process will focus on ensuring the major features of CyberQuest work as intended before the final demonstration.

1.2 Team Members

Resource Name	Role
Brennan	Developer
Brennan/ Brennan's Sisters	Testers
Professor Moore	Stake Holder

Scope

The following functionality is included in the testing scope for the current version of CyberQuest:

User Login

- Username and password fields
- Required fields
- Correct login credentials
- Incorrect login credentials
- Navigation to the dashboard after successful login

Dashboard

- Dashboard loads correctly
- Scenario categories are displayed
- Phishing category navigation
- Suspicious Links category navigation
- Password Security category navigation
- Physical Security category navigation
- Completed scenario count
- Safety score
- Progress bar

Phishing Scenarios

- Phishing Scenario 1
- Phishing Scenario 2
- Selecting suspicious portions of an email
- Submitting answers
- Correct answer scoring
- Incorrect answer scoring
- Feedback pages
- Try Again functionality
- Return to Dashboard functionality

Suspicious Link Scenarios

- Suspicious Link Scenario 1
- Suspicious Link Scenario 2
- Selecting suspicious links
- Submitting answers
- Scoring
- Feedback
- Navigation
- Try Again functionality
- Return to Dashboard functionality

Password Security Scenarios

- Password Scenario 1
- Password Scenario 2
- Answer selection
- Scoring
- Feedback
- Navigation
- Try Again functionality
- Return to Dashboard functionality

Physical Security Scenarios

- Physical Security Scenario 1
- Physical Security Scenario 2
- Answer selection
- Scoring
- Feedback
- Navigation
- Try Again functionality
- Return to Dashboard functionality

Progress Tracking

- Recording completed scenarios
- Recording scenario scores
- Calculating the overall safety score
- Updating the number of completed scenarios
- Updating the progress bar

User Interface

- Page layout
- Buttons
- Navigation

- Readability
- Scenario interaction
- Basic responsive behavior on smaller screens

Out of Scope

The following items are not included in the current testing scope:

- Large-scale load testing with hundreds or thousands of users.
- Production-level database performance.
- Real employee or student account authentication.
- Real email delivery or phishing campaigns.
- Real-world cybersecurity attacks.
- Integration with an organization's existing employee-management system.
- Mobile applications.
- Advanced accessibility testing beyond basic usability and readable interface design.
- Leaderboards or competitive scoring, since CyberQuest is designed around individual progress.
- Features that have not yet been implemented in the current version.

Assumptions / Risks

1.3 Assumptions

1. Users will access CyberQuest through a modern web browser.
2. The application will be run through the project's local development environment during testing.
3. Testers will have access to the CyberQuest project files.
4. Testers will have the necessary software to run the frontend and backend.
5. Users will understand basic computer and web navigation.
6. The scenarios and correct answers are based on the cybersecurity information provided by the project.
7. The number of scenarios may increase as development continues.
8. Testing will be performed throughout development rather than only at the end of the project.

1.4 Risks

The following risks have been identified and the appropriate action identified to mitigate their impact on the project. The impact (or severity) of the risk is based on how the project would be affected if the risk was triggered. The trigger is what milestone or event would cause the risk to become an issue to be dealt with.

#	Risk	Impact	Trigger	Mitigation Plan
1	Changes to the code may break existing features.	High	Errors in code	**Test existing features after major changes.**
2	**Incorrect scoring could give users incorrect feedback.**	High – to schedule and quality	Poor design of feedback navigation	**Test scenarios with correct and incorrect answers.**
3	Broken links could prevent users from navigating the application.	Medium	Navigation links are broken in the code	**Test all navigation buttons and links.**
4	**Limited development time may limit testing.**	Medium	Limited Time	Test core functions first

Test Approach

CyberQuest will primarily use manual testing. Each feature will be tested to make sure it works as expected.

Testing will include:

- Functional testing: Make sure individual features work.
- Integration testing: Make sure features work together, such as completing a scenario and updating the dashboard.
- Usability testing: Make sure users can understand and navigate the application.
- Regression testing: Make sure new changes do not break existing features.

Testing have been performed throughout development so that problems can be found and fixed before the final demonstration.

1.5 Test Automation

No automated testing is planned for the current version of CyberQuest. Testing will be completed manually through the application.

Test Environment

CyberQuest will be tested using:

Frontend: HTML, CSS, and JavaScript
Backend: Python, FastAPI, and Uvicorn
Development: Visual Studio Code and IntelliJ IDEA
Browser: Google Chrome
Testing: Local development environment using Live Server.

Milestones / Deliverables

1.6 Test Schedule

The initial test schedule follows…….

Task Name	Start	Finish	Effort	Comments
Test Planning			1d	
Test login			2 d	
Test dashboard			1 d	
Test all scenarios			2d	Most important
Test scoring and feedback			1d	More work to be done
Test navigation			1d	
Regression testing			1d	
Final testing			9d total	

1.7 Deliverables

Deliverable	For	Date / Milestone
Test Plan	Developer	Defines how CyberQuest will be tested.
Test Cases	Developer	Lists the features and situations being tested.
Test Results	Developer	Records whether tests passed or failed.
Issue List	Developer	Records problems found during testing.
Final Summary	Developer	Summarizes the final testing results.